

ASSIGNMENT 1

Government Engineering College Kishanganj
Department of Computer Science and Engineering
Computer Networks LAB-01

Experiment Title

Study of LAN Topology, Network Devices, and Address Configuration

Problem Statement

Perform the following tasks and prepare a detailed report:

1. Draw the Network Topology:

- Identify and draw the LAN topology of your organization (e.g., Star, Bus, Mesh).
- Clearly label all components such as Router, Switch, Access Point, and end devices (PCs, laptops, etc.).

2. Identify Network Devices:

- List all networking devices used in the LAN.
- Mention the function of each device.

3. Analyze Connection Types:

- Specify whether connections are:
 - Wired (Ethernet)
 - Wireless (Wi-Fi)
- Mention where each type is used.

4. Measure Network Speed:

- Find the link speed of:
 - Wired network (if available)
 - Wireless network
- Mention the tools/commands used (e.g., Task Manager, `netsh`; `netsh wlan show interfaces` or , etc.).

5. Find System Network Details:

- Determine and record the following of your computer:
 - MAC Address
 - IP Address (IPv4)

1

- Subnet Mask
- Mention the command(s) used (e.g., `ipconfig`, `getmac`).

6. Explain Observations:

- Briefly explain:
 - Difference between wired and wireless speed
 - Role of subnet mask in networking

Output

- Network diagram (hand-drawn not software-generated)
- List of devices and their roles
- Screenshots/outputs of commands used
- Table of network parameters (MAC, IP, Subnet Mask)
- Short explanation and conclusion

QUESTION 1.

1. Draw the Network Topology:

- Identify and draw the LAN topology of your organization (e.g., Star, Bus, Mesh).
- Clearly label all components such as Router, Switch, Access Point and end devices (PCs, laptops, etc.)

→ LAN topology of GEC Kishanganj

The College Campus network observed:

It consists of:

- * Layer-3 Core Switches in the Server Room.
- * Managed by PoE Switches.
- * Wireless Access Point (APs)
- * Ethernet port in Classroom & Laboratories.

Network Topology

Hybrid Hierarchical Topology used in my organization combines:

- 1) Tree Topology
- 2) Extended Star Topology
- 3) Ring Topology (Server Room)

→ The network hierarchical layer

- 1) Core Layer
- 2) Distribution Layer
- 3) Access Layer

1) Core Layer

Core layer contains Server Room and Server room have Firewall, Router, 2-L3 Switches, Patch Rack, Server, inventory, etc.

2) Distribution Layer

* Distribution layer contains 4-switches.

* The 4 switches are →

- ① Main Build → R7 (Room no - 213, 2nd Floor)
- ② Main Build → R1 (Room no - 301, 3rd Floor)
- ③ Main Build → R6 (Room no - 113, 1st Floor)
- ④ Main Build → R2 (Room no - 201, 2nd Floor)

* This 4-switches contains →

- ① 18 Port Gigabit Easy Smart Switch with 16-Port.

TL-SG1218 MPE

- ② SG 34 28 X MP → 24 Port Gigabit & 4-port 10 Gb SFP+,
L2+ Managed switches with 24-Port PoE+.

→ Now let's discuss these 2 above components →

1) TP-Link TL-SG1218 MPE

* Likely used for: CCTV Camera, Wi-Fi Access Point, etc.

* 18 Port Gigabit "Easy Smart" Switch.

* 16 Port Support PoE+

2) TP-Link Omada SG 34 28 X MP

* Higher end managed layer 2+ switch.

* 24 PoE+ Port, Uplinks, 10 Gb SFP.

* Usually acts as a main distribution for each floor.

→ Access Layer

* Access layer contain 3-switches i.e

- 1) Main Build R1 [Room no - 301, 3rd Floor]
- 2) Main Build R4 [Room no - 05, Ground Floor]
- 3) Main Build R5 [Room no - 18, Ground Floor]

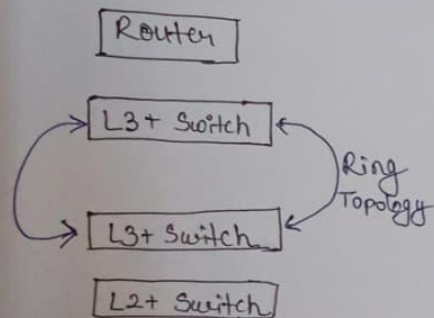
(2)

* Every Switches act as access Layer which helps to provide network connectivity to all classroom, Lab, Faculty Room, etc.

* Access layer directly Connect:

- ① Ethernet Port
- ② Wireless AP
- ③ Lab System
- ④ Outdoor Wireless AP (ex- 018)

→ SERVER ROOM ARCHITECTURE / Topology



Ethernet Port: Port for Wired network connection.

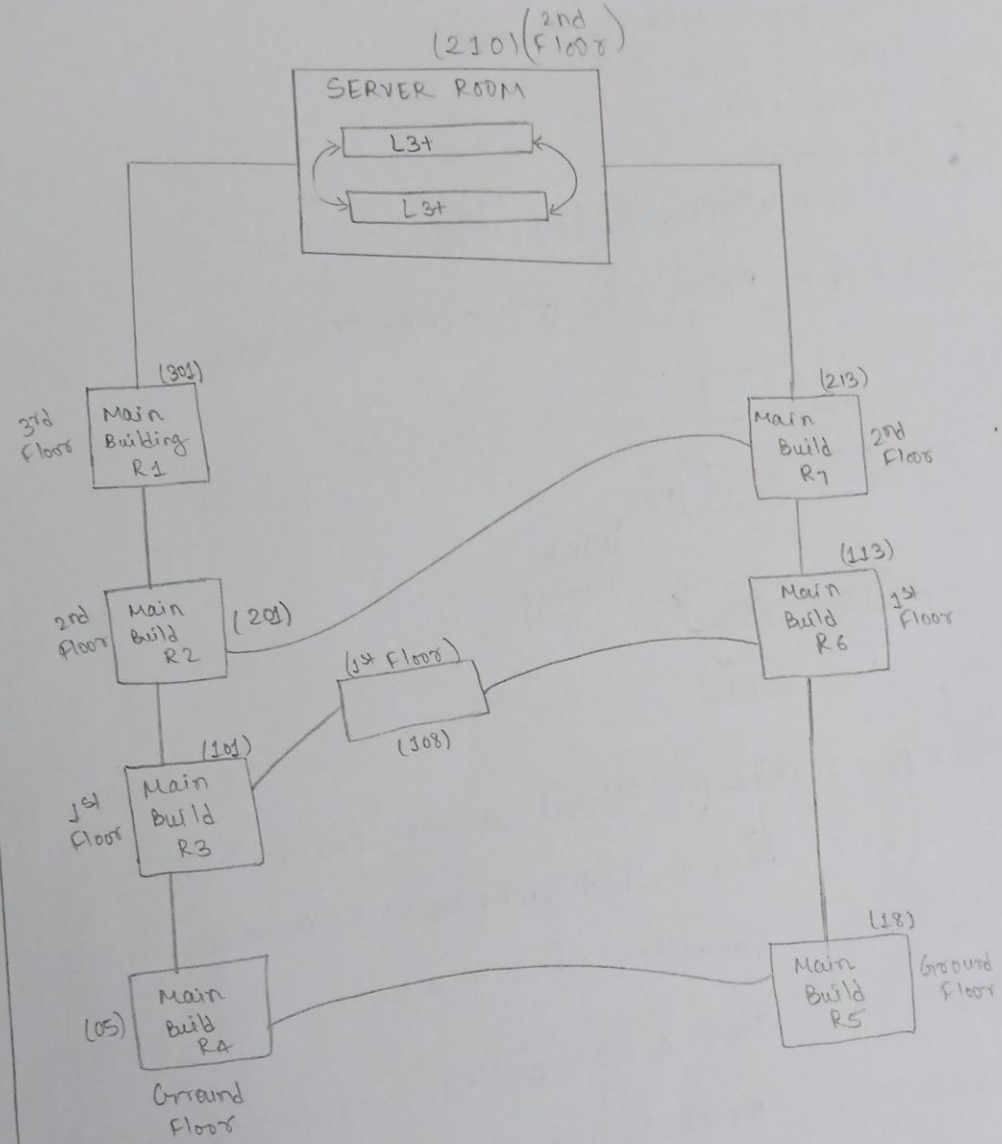
Wireless AP: Device that provides Wi-Fi access.

Lab System: Computer used in a lab.

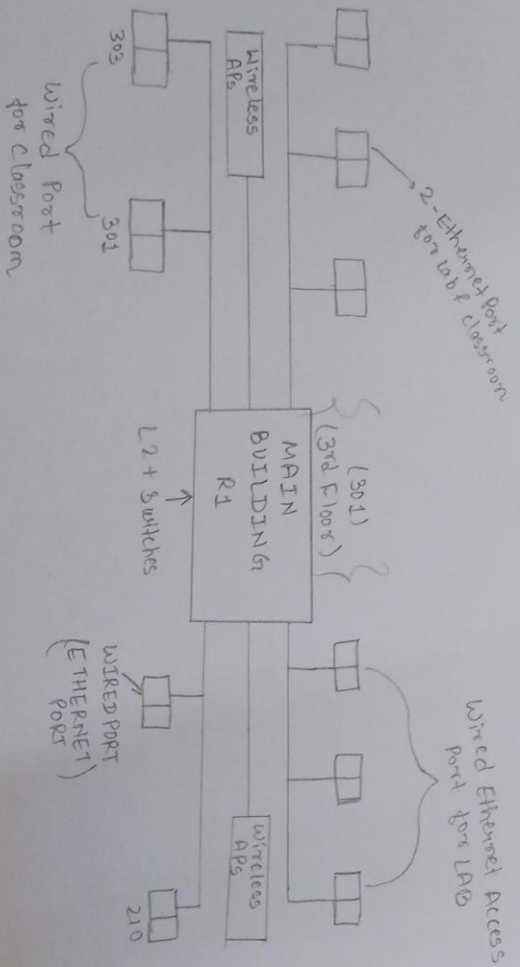
Switch: Device connecting computers in a network.

Ring Topology: Network connected in a circular form.

Topology



Access Layer SWITCH (3rd Floor-301)



QUESTION 2.

2. Identify Network Devices

- List all networking devices used in the LAN
- Mention the function of each device.

1. Router

Function:

- Connects the LAN to the Internet or other networks.
- Assigns IP addresses (DHCP).
- Routes data packets between different networks.

2. Switch

Function:

- Connects multiple devices (PCs, printers, etc.) within the LAN.
- Transfers data to the correct device using MAC addresses.

3. Wireless Access Point / Wi-Fi Router

Function:

- Provides wireless (Wi-Fi) connectivity to devices in the LAN.
- Allows laptops/mobile devices to join the network.

4. Network Interface Card (NIC) / Network Adapter

Function:

- Hardware inside a computer that enables network connection.
- Examples : Wi-Fi Adapter, VMware Network Adapter

5. Computer / Host Device

Function:

- End device used to send, receive, and process data on the LAN.

6. Ethernet Cable

(if wired LAN is used)

Function

- Physically connects devices like computers, switches and routers for data transfer.

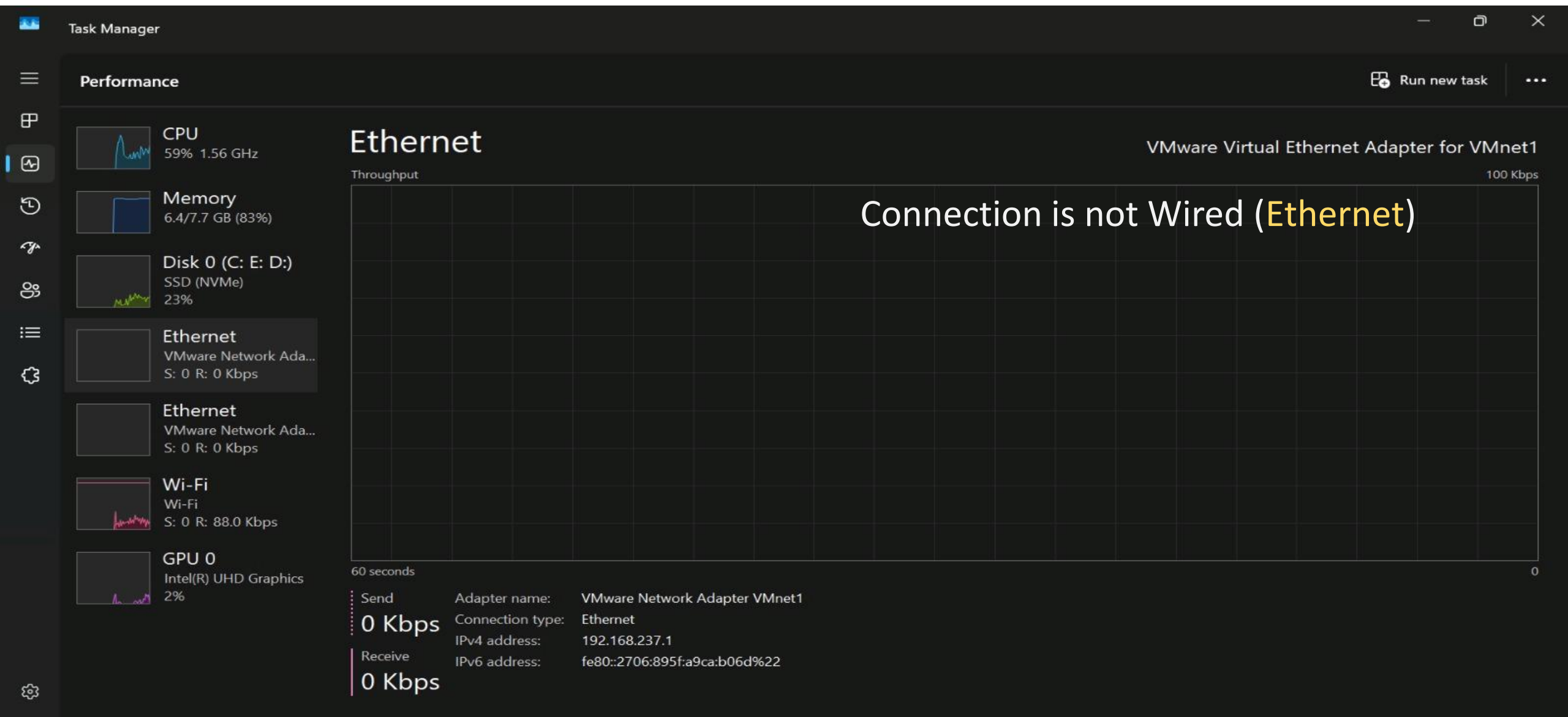
7. Modem

(if Internet access is included)

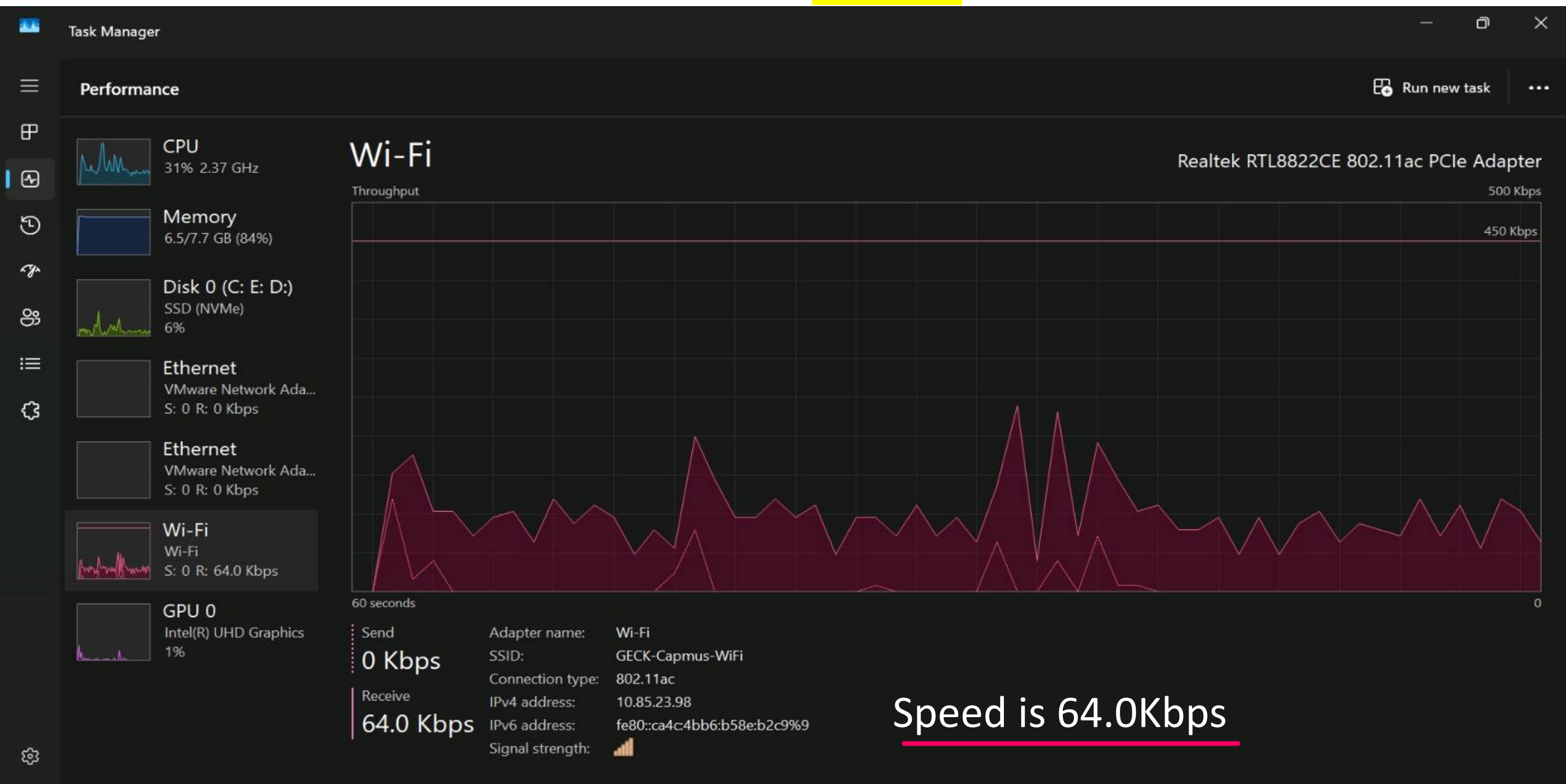
Function:

- Connects the router/LAN to the ISP (Internet Service Provider).
- Converts ISP signals into digital network signals.

Q3. Analyse Connection Types + Q4. Measure Network Speed:



Connection is Wireless (Wi-Fi)



```
Command Prompt
Microsoft Windows [Version 10.0.26100.8117]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>netsh wlan show interfaces or

There is 1 interface on the system:

Name                : Wi-Fi
Description          : Realtek RTL8822CE 802.11ac PCIe Adapter
GUID                 : 3b0c1f56-09b4-443a-ad6a-4e47aaf8cbee
Physical address     : 5c:ba:ef:8e:5f:75
Interface type       : Primary
State                : connected
SSID                 : GECK-Capmus-WiFi
AP BSSID             : b0:19:21:74:04:91
Band                 : 5 GHz
Channel              : 153
Connected Akm-cipher : [ akm = 00-0f-ac:02, cipher = 00-0f-ac:04 ]
Network type         : Infrastructure
Radio type           : 802.11ac
Authentication       : WPA2-Personal
Cipher               : CCMP
Connection mode       : Auto Connect
Receive rate (Mbps)  : 780
Transmit rate (Mbps) : 780
Signal               : 100%
Rssi                 : -50
Profile              : GECK-Capmus-WiFi
QoS MSCS Configured  : 0
QoS Map Configured   : 0
QoS Map Allowed by Policy : 0
```

Use of netsh wlan show interfaces or

```
Command Prompt
Microsoft Windows [Version 10.0.26100.8117]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>getmac

Physical Address      Transport Name
=====
00-50-56-C0-00-01    \Device\Tcpip_{E6953589-EEF3-4362-8C55-0465FD871FBA}
00-50-56-C0-00-08    \Device\Tcpip_{6C6C975F-A107-4A33-A568-634B787F9C14}
5C-BA-EF-8E-5F-75    \Device\Tcpip_{3B0C1F56-09B4-443A-AD6A-4E47AAF8CBEE}

C:\Users\HP>
```

Use of getmac

5. Find System Network Details:

```
Windows PowerShell
PS C:\Users\HP> ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 12:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::2706:895f:a9ca:b06d%22
    IPv4 Address. . . . . : 192.168.237.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::9c5c:8fd6:c3b0:6725%12
    IPv4 Address. . . . . : 192.168.107.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::ca4c:4bb6:b58e:b2c9%9
    IPv4 Address. . . . . : 10.85.23.98
    Subnet Mask . . . . . : 255.255.248.0
```


Question 6.

6. Explain Observations

- Briefly explain:
 - Difference between wired and wireless speed
- Wired Speed - Data transfer through cables (LAN/Ethernet)
- Wireless Speed - Data transfer through Wi-Fi/radio waves

Wired Speed	Wireless Speed
① Uses Cable (Ethernet/LAN) connection	① Uses Wi-Fi/radio waves.
② Usually faster and more stable.	② Usually slower compared to wired
③ Less interference	③ Can be affected by walls, distance, devices
④ Lower latency (better for gaming/video calls)	④ Higher latency
⑤ Example: 100Mbps - 1Gbps+	⑤ Example: 20Mbps - 600Mbps+ (depends on Wi-Fi type)
⑥ Faster, stable, secure	⑥ Convenient, portable, but speed may vary.

— Role of Subnet mask in networking

A Subnet Mask is used to divide an IP address into Network ID and Host ID.

Functions of Subnet Mask:

1. Identifies the network part of an IP address.
2. Identifies the host/device part of an IP address.
3. Helps devices determine whether another device is on the same network or different network.
4. Reduces network traffic by creating subnets (smaller networks).

Example:

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

- Network ID → 192.168.1.0
- Host ID → 10

Network Diagram

